Rev0 Maxwell Technical Report

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Rev0 (one 5-board PCB stator, one 16-magnet rotor) has a full-motor Maxwell torque constant of 18.53 mN·m/A. That is 27.8 mN·m per amp peak and 1.37 V rms phase back-EMF per 1000 rpm. Three independent methods agree on it within 0.03–5 %, and the rotating model matches the static results to 1.5 %.

Machine and simulation chain

Rev0 is a single-sided coreless axial-flux machine: five 6-layer PCBs stacked as one stator, facing one rotor of 16 magnets on a steel backplate.

Item	Value used in the models
Stator	5 boards, 6 layers each, 1 oz copper, 1.0 mm FR-4, board pitch 1.02 mm; 3 phases, 8 coils per phase per board
Rotor	16 × N35, 20 × 6 × 2 mm, centred at 28 mm radius, alternating polarity (8 pole pairs)
Backplate	2 mm steel 1010, OD 86 / ID 19 mm (hardware is 80 / 25 mm: correction pending)
Air gap	1.0 mm, magnet face to board-1 surface
Shaft	19 mm steel
Reference speed	1000 rpm = 133.3 Hz electrical

The results come from five Ansys Maxwell 2025 R2 models, each checking the next:

1. **Board-by-board detailed FEM:** each board solved alone with its real traces and vias, then summed (lab PCs).
2. **Field map:** magnets and steel only, with B sampled on the trace centre-lines and Lorentz force integrated (lab PCs).
3. **Full-motor detailed static:** all five boards meshed together, all three phases at 1 A synchronous, two rotor angles, 6 adaptive passes, 110 GB per angle (UCT HPC, 40 cores).
4. **Rotating field-only transient (FMROT):** the rotor turns at 1000 rpm through one electrical period in 24 steps, with B exported on the traces at every step (UCT HPC).

5. **Equivalent-coil model (EQ):** each phase per board is replaced by 8 stranded sector coils calibrated to model 3; 8 rotor angles (UCT HPC).

The torque constant K is defined by $T = 1.5 \cdot K \cdot I_{pk}$. The phase back-EMF peak equals $K \cdot \omega_{mech}$.

Torque constant

The full-motor solve gives K = 18.534 mN·m/A from the copper (Lorentz) torque. This matches the sum of the single-board solves to 0.03 %. The field map reads 4.9 % high.

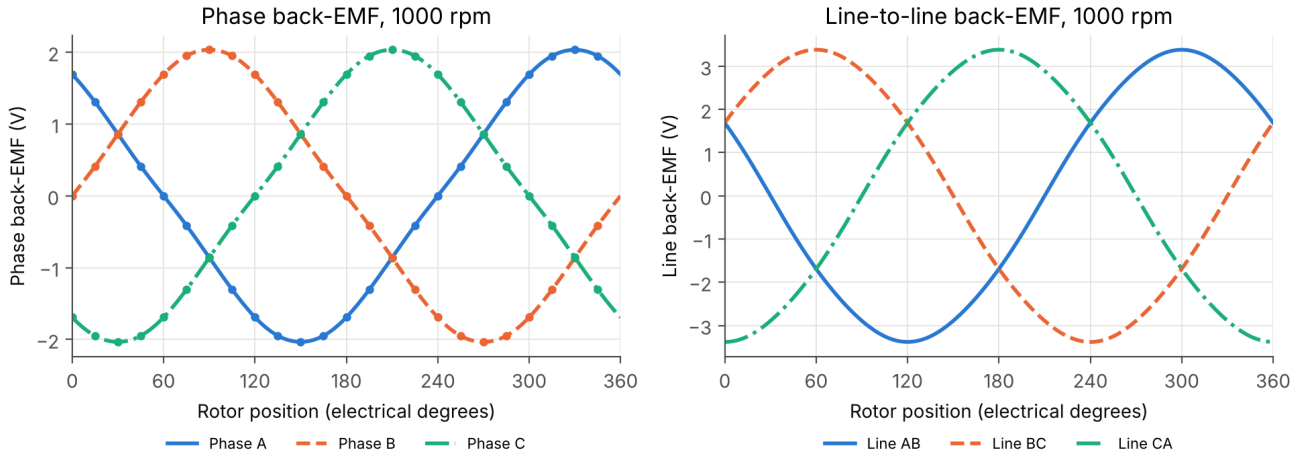
Method	K (mN·m/A)	vs full motor
Full-motor static, copper torque (2-angle mean)	18.534	reference
Full-motor static, rotor virtual work	18.470	−0.3 %
Board-by-board detailed sum	18.540	+0.03 %
Field map (centre-line Lorentz)	19.496	+5.2 %
Rotating field-only transient	19.475	+5.1 % (same basis as the field map)

Board (1 = next to the magnets)	1	2	3	4	5
Full motor, K (mN·m/A)	6.282	4.590	3.364	2.474	1.825
Field map vs full motor	+1.4 %	+4.5 %	+6.7 %	+10.2 %	+10.4 %

- Copper and rotor torque agree within 0.51 % (0°) and 0.18 % (3.75°), which confirms the mesh.
- K falls by about 26 % per board: the field decays at 0.30 per mm through the stack.
- The field map's excess grows with distance from the magnets. The likely cause is that centre-line sampling differs from the volume force on the meshed copper. Mesh, padding and shaft stub were each ruled out.
- Torque is linear in current: 2 A gives exactly twice the torque of 1 A, within 0.002 %.
- Results from the field map and the rotating model are scaled by 18.534 / 19.475 = 0.952 to the full-motor value.

Back-EMF, torque ripple and cogging

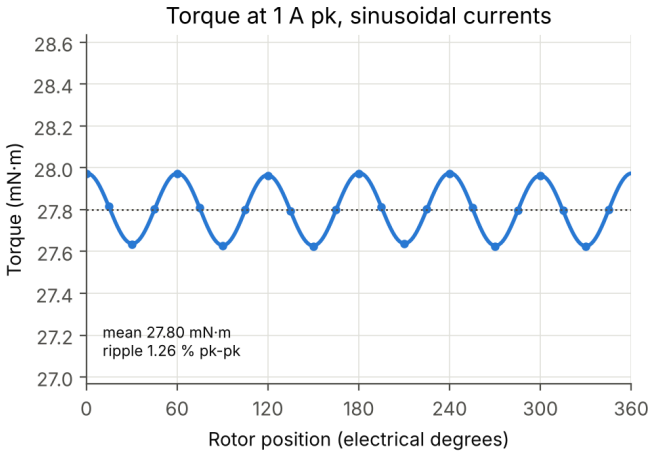
At 1000 rpm the phase back-EMF is 1.372 V rms and the line back-EMF is 2.377 V rms. The line waveform is close to sinusoidal at 0.7 % THD, because the phase waveform's 3rd harmonic cancels between phases.

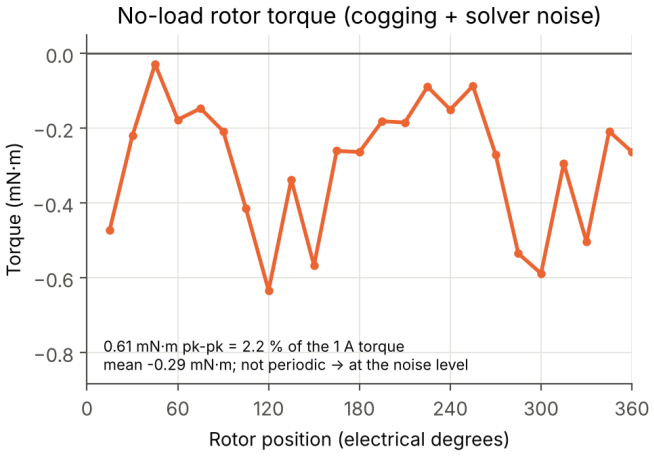


Scaled to the full-motor torque constant (K = 18.53 mN·m/A). Points: Maxwell time steps; lines: Fourier fit to the 11th harmonic. Phase 1.372 V rms, THD 5.5 %; line 2.377 V rms, THD 0.7 %.

Quantity	Phase	Line
Fundamental at 1000 rpm	1.372 V rms (1.941 V pk)	2.377 V rms
Back-EMF constant	1.372 V rms per 1000 rpm	2.377 V rms per 1000 rpm
THD	5.5 % (H3 5.5, H5 0.7, H7 0.05 %)	0.7 %
Phase balance	K spread 0.04 %; displacement 120.0°	—

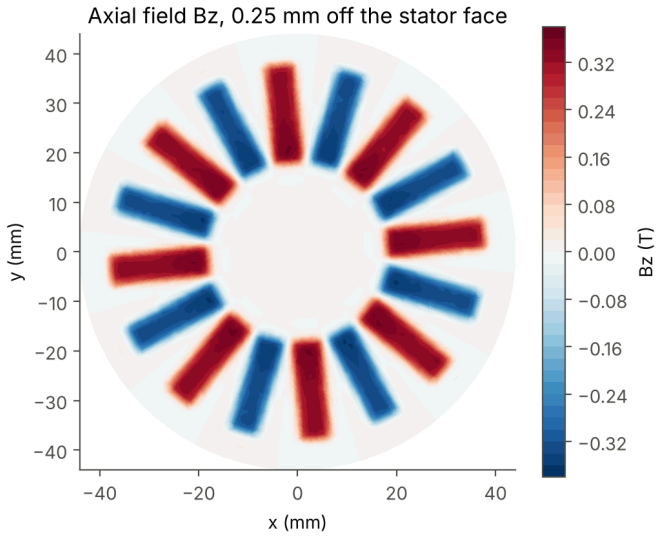
- The rotating model agrees with the static field map within 1.49 % of K1 at all 120 board-phase-angle points, with board 1 within 0.4 % and 0.2°. Rotation adds no measurable effect: there are no eddy currents in the magnets (no armature field) and none in the steel (it turns with the magnets).
- With sinusoidal currents at 1 A pk, mean torque is 27.80 mN·m with 1.26 % pk-pk ripple at 6× electrical frequency, set by the 5th and 7th back-EMF harmonics.
- The no-load rotor torque varies 0.61 mN·m pk-pk but is not periodic. The static board solves gave the same picture (below 0.47 mN·m, not periodic). Cogging is therefore below the solver noise floor, as expected with no stator iron.

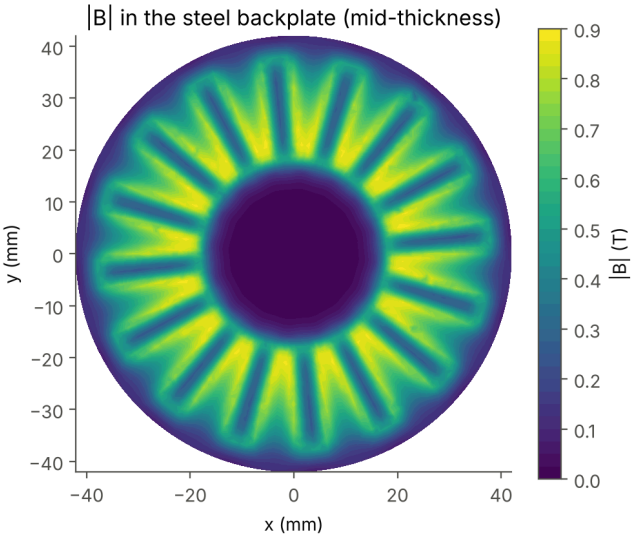




Field distribution and current density

The air-gap field peaks at 0.44 T just off the stator face, and the backplate peaks at 0.90–0.92 T. That is well below the 1.5–1.6 T knee of 1010 steel, so the 2 mm plate is not saturated.

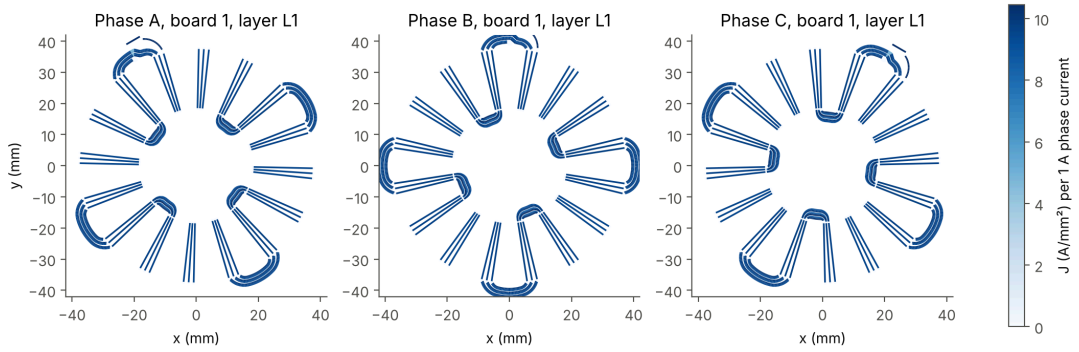




Rev0 rotating model at t = 0.9375 ms (1000 rpm). Gap |B| max 0.443 T; steel |B| max 0.897 T (no point above 1.6 T).

Quantity	Value	Source
Bz 0.25 mm off the stator face	max 0.448 T, mean	B
Fundamental Bz on the tracks, board 1 / board 5	0.174 / 0.051 T	Field map
Field decay through the stack	0.30 per mm (26 % lost per board)	Field map
Backplate	B	, max / 99th percentile / mean
Backplate	B	max, static

Current density per 1 A of phase current is 9.2–9.4 A/mm² in the tracks, with a whole-board mean of 8.6 A/mm² and a 95th percentile of 10.3 A/mm². Each phase splits over parallel tracks on all six layers. Local peaks reach 57 A/mm² per A at narrow necks near the terminals and vias; these are short hot-spot candidates but carry little of the total loss. Skin depth at 1000 rpm is 5.7 mm (164× the copper thickness), so the distribution is effectively DC.



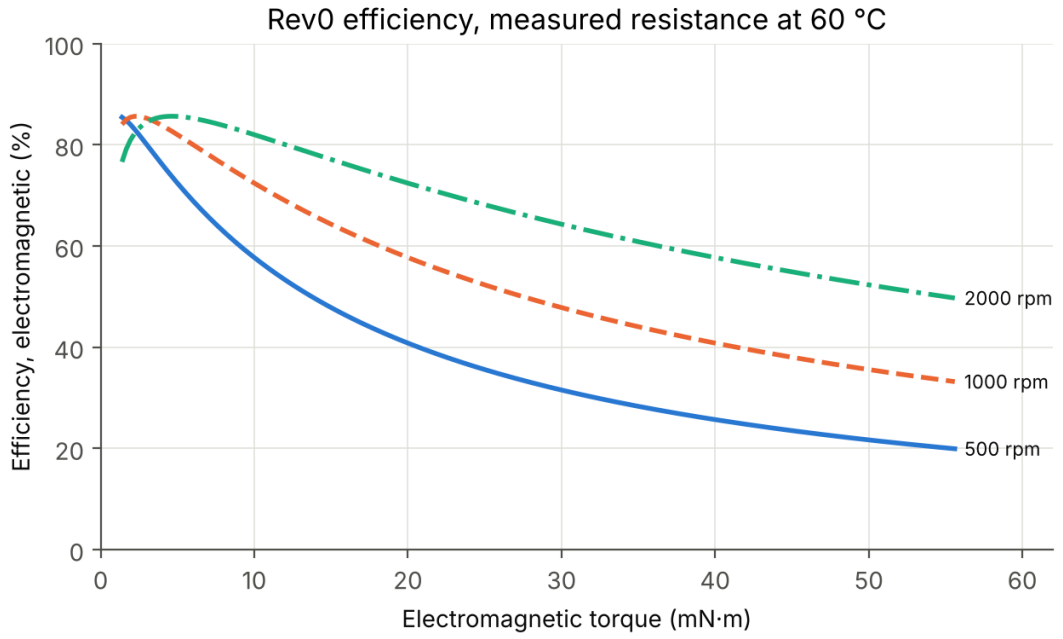
DC current share from the nodal solve over a 0.5 mm × 35 µm track, all six layers similar (median 9.2–9.4 A/mm² per A). Whole board: mean 8.6, 95th percentile 10.3, max 57.1 A/mm² per A (local necks; colour scale clipped at the 99.5th percentile).

Losses, efficiency and thermal limit

Copper I²R loss dominates: 2.9 W at 1 A pk against 20 mW of track eddy loss at 1000 rpm. Electromagnetic efficiency at the 1000 rpm, 1 A pk test point is therefore about 50 %, rising with speed and falling with current.

Loss	Value	Notes
Phase resistance at 20 °C	1.54–1.83 Ω measured (5 × 0.32–0.34 Ω + interconnect); model 1.33–1.39 Ω	Measured value used below

Loss	Value	Notes
Copper loss at 1 A pk, 60 °C	2.93 W (1.5 · I² · R, R = 1.95 Ω)	Scales with I² and winding temperature
Track eddy loss at 1000 rpm	20.4 mW (boards 1–5: 10.8 / 4.8 / 2.5 / 1.5 / 0.8 mW)	Thin-strip estimate, ∝ speed²; 0.33 W at 4000 rpm
Magnet and steel eddy loss	0	No armature field in the model; steel turns with the magnets
Phase inductance	about 110 µH (analytic, ±50 %)	ωL = 0.09 Ω at 1000 rpm, negligible against R



R = 1.95 Ω per phase (mid of measured 1.54–1.83 Ω at 20 °C, at 60 °C); copper and track-eddy loss only — bearing and windage loss not included.

Speed	0.5 A pk (13.9 mN·m)	1.0 A pk (27.8 mN·m)	1.5 A pk (41.7 mN·m)
500 rpm	49.7 %	33.2 %	24.9 %
1000 rpm	65.9 %	49.7 %	39.8 %
2000 rpm	78.2 %	65.9 %	56.7 %

These efficiencies are electromagnetic only. Bearing and windage loss are not modelled, and they will lower the measured shaft efficiency at these small power levels.

Thermal limit: the earlier Icepak model gave 8.75 K/W from the hottest board to air, assuming an acrylic hood with fans and aluminium end plates. With the measured resistance, the provisional continuous limit is **0.85 A pk** (about 24 mN·m). That holds at a

60 °C board limit with a pessimistic 15 K/W, pending confirmation of the holder material and the as-built cooling. 1.0 A pk is a supervised test point. N35 magnets are limited to 80 °C.

Equivalent-coil model, limitations and open items

The equivalent-coil model reproduces the full-motor result once calibrated, but its turn count is not yet predictive. Each phase per board is represented by 8 stranded sector coils on two layer slabs, which preserves the angular and radial position of the coil sides, the board heights and the phase polarity. With geometric turns (Ncoil = 2.006) it gives K ≈ 23.0 mN·m/A, 24 % high. One factor of 0.809 (Ncoil = 1.623) matches the full-motor value. After that:

- every board agrees within about ±4 % (spread 7.1 %);
- the phases agree within 1.1 %;
- the phase angle is within 0.4° and the phase displacement is 120.0°.

Copper resistance and loss are handled outside the field model, with no bulk conducting block.

Limitations:

- **Geometry:** the backplate is modelled at OD 86 / ID 19 mm, but the hardware is 80 / 25 mm. A board-1 check run is still to be done; the effect is expected to be within a few percent.
- **As-built gaps:** the air gap and inter-board gaps are unmeasured. K drops about 7 % per extra 0.25 mm of air gap and about 4 % per 0.1 mm of inter-board gap.
- **Rotating model scope:** armature reaction is neglected. That is valid for a coreless stator, where the armature field is in mT against a 0.2–0.4 T magnet field.
- **Eddy loss:** the estimate samples B at track mid-points and may be 10–20 % high. Pads, pours and vias are not included.
- **Equivalent-coil turns:** the 24 % geometric over-prediction is not yet explained, so report it as a calibrated model.
- **Thermal:** the limits rest on an assumed housing until the assembly is characterised.

Pending: the backplate 80 / 25 mm correction run, a one-step mesh refinement of the equivalent-coil model, and measured back-EMF, resistance and temperature rise from the test bench to compare against this report.